Review Questions for Lecture 17

1. Explain why it would be difficult, if not impossible, to take a simple random sample of all Americans.

2. What is the key difference between a quota sample and a stratified sample?

3. What is the key difference between a cluster sample and a stratified sample?

4. Describe briefly the process of stratified sampling.

5. Describe briefly the process of multi-stage cluster sampling.

6. In a typical poll, is the greater concern bias or chance error? Explain briefly.

7. All else equal (e.g. sample size), which typically results in a more representative sample: a simple random sample or a stratified sample? Explain briefly why.

8. All else equal (e.g. sample size), which typically results in a more representative sample: a simple random sample or a cluster sample? Explain briefly why.

9. What are two different methods to randomly sample people by telephone? For each method, discuss briefly the chief difficulty of the method.

10. The following example uses simulation to investigate simple random sampling, cluster sampling, and stratified sampling. Suppose that you are interested in estimating the mean height in a hypothetical country. This country consists of 3 groups, with each group having $1{,}000{,}000$ people. The 3 groups have mean heights of 68,70, and 72 inches, and the standard deviation of heights is 3 inches in each group. The code below creates heights for this hypothetical country.

```
set.seed(12345)
group1 = rnorm(1000000,68,3)
group2 = rnorm(1000000,70,3)
group3 = rnorm(1000000,72,3)
country=c(group1,group2,group3)
```

   a. Use simulation to estimate the standard error for a simple random sample of 150 people from the country. Run at least $10{,}000$ simulations. Note that you will want to use the option replace=FALSE in the sample command.

   b. Use simulation to estimate the standard error for a cluster sample of 150 people from the country. You will first want to randomly select the group that you are sampling from, with each group having a $1/3$ chance of being chosen. You will then want to perform a simple random sample of 150 people from that group. Run at least $10{,}000$ simulations.

c. Use simulation to estimate the standard error for a stratified sample of 150 people from the country, with 50 people sampled from each group. Run at least $10{,}000$ simulations.

d. Compare the results from parts a,b, and c.